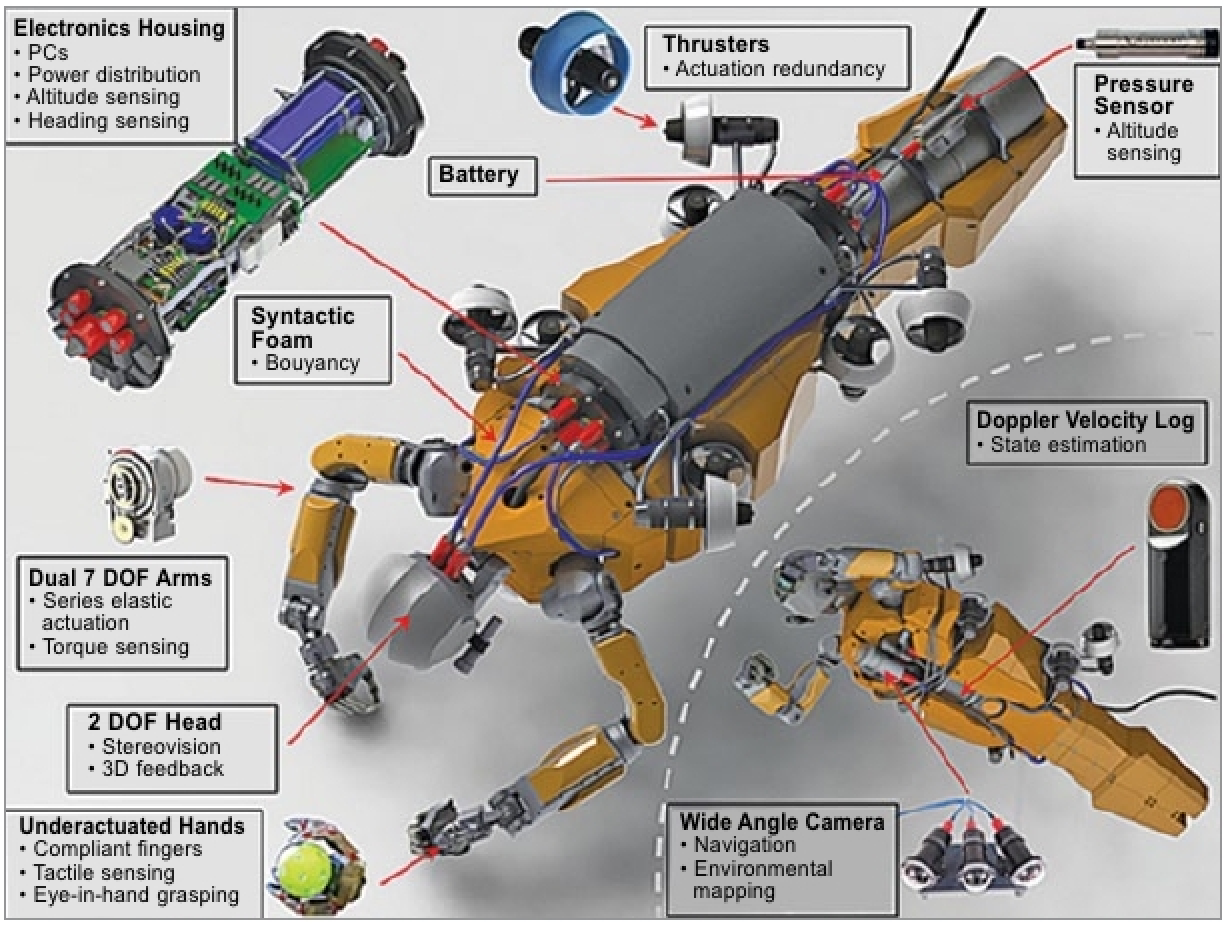

Fig. 1: OceanOne ROV for deep-sea exploration (Credit: www.dogonews.com)

tion and data management. Various programs, which the team developed, have ensured that the data is brought together and then unusable image material, such as blurred images, is removed. Researchers then use an AI algorithm that automatically records the data for evaluation. The method can be transferred to any project, including autonomous underwater vehicles (AUVs) or camera systems.

AI and robots for exploration

Researchers have demonstrated how the use of autonomous robotics and AI at sea can dramatically accelerate the exploration and study of hard-to-reach deep-sea ecosystems. Use of AI and machine learning could help identify the links between the world's oceans. Researchers from Massachusetts Institute of Technology's (MIT) Department of Earth, Atmospheric and Planetary Sciences have used computers to reveal connections and patterns in the oceans, which would otherwise be impossible to achieve by human beings.

An autonomous robot for ocean application is something like a self-driving car but in the ocean. Researchers have deployed a number of AUVs with high-altitude 3D visual mapping cameras on seafloors. Seafloor images are gathered and algorithmically investigated to discover biological hotspots. Researchers then target them for interactive sampling and observations. Algorithms used are able to rapidly produce simple summaries of observations and form subsequent deployment plans.

This project has demonstrated how modern data science can greatly increase the efficiency of conventional research at sea and improve the productivity of interactive seafloor exploration. Using unsupervised clustering algorithms, researchers have identified dynamic hotspots in image data for more detailed surveys and sampling by a remotely-operated vehicle (ROV).

Nereus is the world's deepest-diving underwater vehicle. It can be configured to operate as an ROV, or operate independently of human control as an AUV. OceanOne ROV developed by Stanford Robotics Lab is shown in Fig. 1.

Oil and gas exploration

AI has a number of potential applications in the oil and gas industry, from surveying, planning, forecasting and facility management to safety. It has helped precision-drilling and automated inspections, improved efficiency of various operations and enabled informed decision-making. Industry giants are already deploying AI in oil and gas business operations. AI is the key to unlocking the next productivity revolution in the industry.

ExxonMobil, USA, is working with MIT to design AI robots for ocean exploration. There is also a report that Gazprom and Yandex (Russia's leading Internet company) has entered into a co-operation agreement for the implementation of new projects in the oil and gas industry.

In India, companies like Hindustan Petroleum Corp. Ltd (HPCL) are using analytics for management dashboards, reporting and decision-making and data visualisation, and geo maps for sales analytics.

Crabster (Fig. 2), developed by Korea Institute of Ocean Science and Technology, is the world's deepest

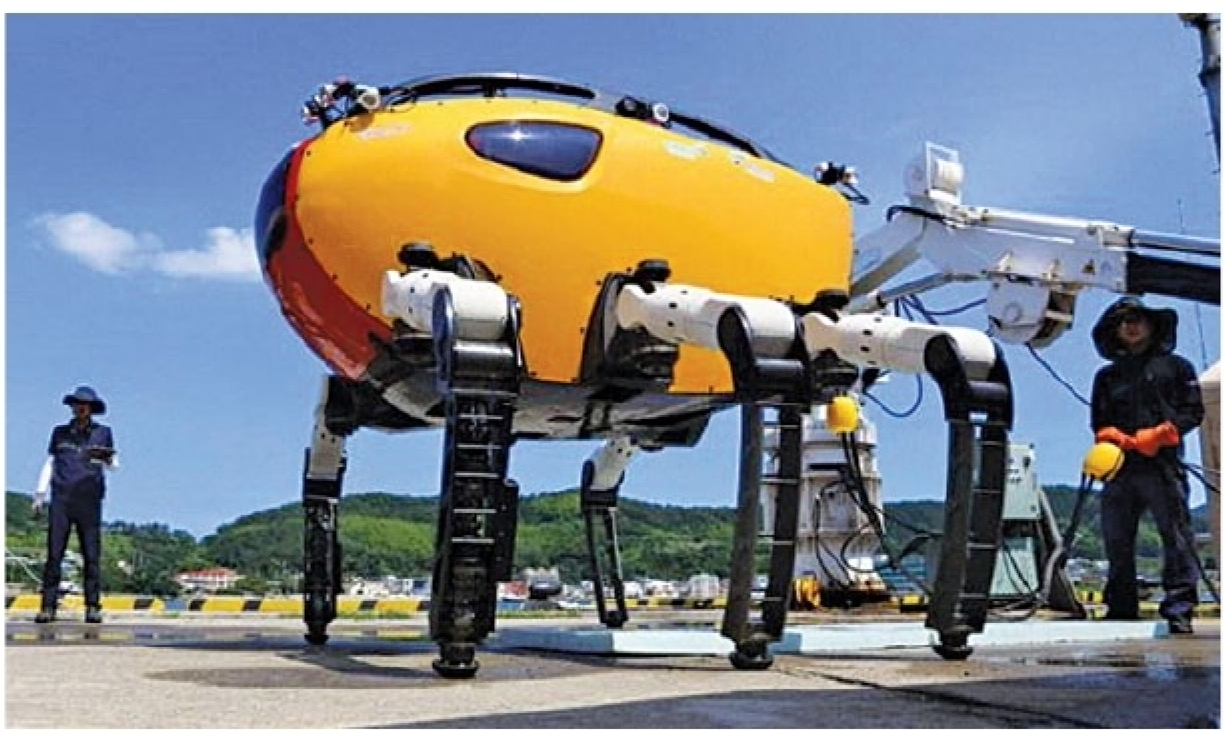

Fig. 2: Crabster robot for fixing structures underwater (www.dailymail.co.uk)